

ENGINEERING ROADMAP

2-YEAR COMPLETE PROGRAM

A measured quadruped, balanced hexapod and professional robotics portfolio

STUDY LOAD

20-hour baseline: 2 hours Monday-Friday + 10 practical hours Sunday; 22 distributed one-hour recovery sessions

TOTAL BUDGET

1,940 hours: 24 x 75-hour modules + four 35-hour integration gates

CALENDAR

12 October 2026 - 22 December 2028 with all declared 2026-2028 vacations

PRIMARY OUTCOME

Junior robotics engineering readiness plus one defensible physical robot release

SPECIALIZATION

Legged robotics, active body leveling, safe AI assistance and measurable validation

Claim boundary

The program targets practical understanding and one verified implementation per common robotics domain. It does not claim research-level mastery, an accredited degree, dynamic running, jumping, or production certification.

PROGRAM CONTRACT AND REALISTIC OUTCOME

WHAT 1,940 HOURS CAN ACCOMPLISH

- Working understanding of all common robotics domains plus practical evidence in each.
- Deeper specialization in ROS 2, embedded integration, legged estimation/control and physical validation.
- A slow 12-DOF quadruped is the mandatory physical capstone; the 18-DOF hexapod is a gated second-platform scale-up.
- AI is used for perception or bounded policy assistance only after classical control and safety are stable.
- The program supports junior robotics applications; professional experience is still required for senior or research roles.

BUDGET ARITHMETIC

Component	Count	Hours	Purpose
Core modules	24 x 75 hours	1,800	Concepts, labs and verified artifacts.
Integration gates	4 x 35 hours	140	Recovery, reviews, rework and cross-domain integration.
Total	97 active weeks	1,940	20-hour baseline plus 22 distributed recovery hours.

Important calendar effect

The fixed start is 12 October 2026 and the finish is 22 December 2028. The new 30 September-8 October 2028 vacation creates a 22-hour shortfall at the normal rhythm, so 22 active weeks receive one extra theory/review hour. Average load is about 20.2 hours per active week; all vacations remain zero-load.

DATED CURRICULUM SCHEDULE 1/2

Block	Focus	Start	Finish	Hours
M1	Engineering workflow: Python, modern C++ and Rust literacy	12 Oct 2026	08 Nov 2026	75
M2	Applied mathematics I: linear algebra, calculus and ODEs	08 Nov 2026	03 Dec 2026	75
M3	Probability, statistics and numerical optimization	04 Dec 2026	10 Jan 2027	75
M4	Electronics and signal processing from zero	10 Jan 2027	05 Feb 2027	75
M5	CAD, mechanics, materials and manufacturing	07 Feb 2027	02 Mar 2027	75
M6	Embedded systems, buses, real-time behavior and safety	03 Mar 2027	28 Mar 2027	75
G1	Foundation and safety integration	28 Mar 2027	11 Apr 2027	35
M7	Actuators, motor control and system identification	11 Apr 2027	16 May 2027	75
M8	Robot geometry and kinematics	17 May 2027	13 Jun 2027	75
M9	Rigid-body dynamics and physical modelling	13 Jun 2027	30 Jul 2027	75
M10	Classical, digital and frequency-domain control	01 Aug 2027	05 Sep 2027	75
M11	State-space control, estimation and constrained optimization	05 Sep 2027	10 Oct 2027	75
M12	ROS 2 production software and observability	10 Oct 2027	07 Nov 2027	75
G2	Model, control and ROS architecture integration	07 Nov 2027	18 Nov 2027	35

Weekly rhythm

Monday-Friday: 2 hours/day, primarily theory, derivation, code review and small exercises. Sunday: 10 practical hours. One evenly distributed Monday in 22 active weeks receives a third hour for review or documentation; Saturdays remain unscheduled.

DATED CURRICULUM SCHEDULE 2/2

Block	Focus	Start	Finish	Hours
M13	Simulation, digital twins and HIL	18 Nov 2027	13 Dec 2027	75
M14	Computer vision and 3D perception	13 Dec 2027	20 Jan 2028	75
M15	Localization, mapping and SLAM	20 Jan 2028	14 Feb 2028	75
M16	Motion planning, decision making and autonomy	15 Feb 2028	22 Mar 2028	75
M17	Manipulation, HRI, ethics, security and industrial breadth	23 Mar 2028	16 Apr 2028	75
M18	Machine-learning foundations for robotics	17 Apr 2028	24 May 2028	75
G3	Autonomy, perception and AI integration	25 May 2028	05 Jun 2028	35
M19	Legged robotics: estimation, gait and footholds	05 Jun 2028	12 Jul 2028	75
M20	Advanced legged control and whole-body methods	12 Jul 2028	06 Aug 2028	75
M21	Robot learning: imitation, RL and sim-to-real	06 Aug 2028	12 Sep 2028	75
M22	Physical quadruped integration and calibration	12 Sep 2028	19 Oct 2028	75
M23	Hexapod scaling, terrain and reliability	19 Oct 2028	13 Nov 2028	75
M24	Validation, systems engineering and professional release	13 Nov 2028	10 Dec 2028	75
G4	Final validation and external defense	10 Dec 2028	22 Dec 2028	35

Schedule assumption

All vacations supplied through 31 December 2028 are included. Later dates are provisional and move by the exact hours lost to future leave, illness or major hardware delays.

DECLARED VACATIONS AND CALENDAR RULES

Vacation start	Vacation end	Rule
24 Dec 2026	03 Jan 2027	Zero study load; unfinished work moves forward.
01 May 2027	10 May 2027	Zero study load; unfinished work moves forward.
19 Jun 2027	28 Jun 2027	Zero study load; unfinished work moves forward.
10 Jul 2027	18 Jul 2027	Zero study load; unfinished work moves forward.
21 Aug 2027	29 Aug 2027	Zero study load; unfinished work moves forward.
25 Sep 2027	03 Oct 2027	Zero study load; unfinished work moves forward.
25 Dec 2027	03 Jan 2028	Zero study load; unfinished work moves forward.
04 Mar 2028	12 Mar 2028	Zero study load; unfinished work moves forward.
29 Apr 2028	08 May 2028	Zero study load; unfinished work moves forward.
24 Jun 2028	02 Jul 2028	Zero study load; unfinished work moves forward.
19 Aug 2028	27 Aug 2028	Zero study load; unfinished work moves forward.
30 Sep 2028	08 Oct 2028	Zero study load; unfinished work moves forward.
23 Dec 2028	31 Dec 2028	Zero study load; unfinished work moves forward.

RECOVERY RULES

- Never double Sunday hours to compensate for a vacation.
- A failed gate consumes gate time first; it does not silently remove later fundamentals.
- Hardware delivery delays trigger simulation, replay or documentation work from the same module.
- If sustained capacity falls below 20 hours/week, preserve module hours and move the finish date.

SOFTWARE ENVIRONMENT ARCHITECTURE

Layer	Primary environment	Purpose	Rule
Mac host	macOS on MacBook Pro M4	Editing, Git, CAD, slicing, MCU tools and documentation.	Install native Apple-silicon builds where available.
Robotics workstation	Ubuntu 24.04 ARM64 in UTM	ROS 2 Jazzy, Gazebo Harmonic, RViz, perception and autonomy labs.	Keep a restorable VM snapshot and lock project dependencies.
Robot computer	Ubuntu-capable Pi 5 or x86	Onboard ROS graph, drivers, estimation, control and selected perception.	Choose after profiling; validate ARM64 package availability.
Reproducible services	Docker images and CI	Headless tests, builds and data-processing jobs.	Do not rely on containers alone for USB/GUI/real-time hardware work.
Accelerated learning	NVIDIA Linux machine or cloud	Isaac Lab/CUDA and learned locomotion only if approved at G3.	The Mac M4 is not the target Isaac/CUDA platform.

INSTALLATION ACCEPTANCE CHECKS

- Git clone/commit/push works; CMake builds a C++ test; Python creates a locked virtual environment.
- The Ubuntu VM has working graphics, USB forwarding, network access, snapshots and at least 80 GB free disk.
- ROS 2 talker/listener, RViz and rosbag record/playback pass before any custom workspace is added.
- Gazebo Harmonic opens a world and gz_ros2_control drives a test joint without version mixing.
- ESP32 compiles, flashes and produces timestamped serial output; logic software decodes one real bus.
- CAD exports a dimensioned drawing and STL; Creality Print slices a calibration part for the CR-10 SE.

SOFTWARE INSTALLATION CHECKLIST 1/4: MAC HOST AND REPRODUCIBILITY

When	Environment	Software	Status	Purpose / decision rule	Link
Now	macOS / host	Homebrew	Required	Package manager for command-line development tools.	Open
Now	macOS / host	Git	Required	Version control for code, CAD exports, configuration and reports.	Open
Now	macOS / host	Visual Studio Code - Apple silicon	Required	Primary editor; add C/C++, Python, CMake and Remote extensions.	Open
Now	macOS / host	UTM	Required	Run an Ubuntu 24.04 ARM64 virtual machine on the M4.	Open
Now	macOS / host	Docker Desktop - Apple silicon	Recommended	Reproducible CI and headless services; not the only ROS hardware environment.	Open
M1	macOS + Ubuntu	CMake	Required	C/C++ build system used throughout the roadmap.	Open
M1	macOS + Ubuntu	uv + Python 3	Required	Isolated Python environments and locked dependencies.	Open
M1	macOS + Ubuntu	Rustup, Cargo and rustfmt	Required literacy	Safe systems-language practice; Python/C++ remain primary for ROS interoperability.	Open
M1	Ubuntu VM	GCC/G++, GDB and clang-format	Required	Native compilation, debugging and consistent formatting.	Open

Installation policy

Install only the software needed for the current module. Record exact versions and setup notes in the repository. Prefer official installers and documentation; do not mix ROS or Gazebo distributions inside one workspace.

SOFTWARE INSTALLATION CHECKLIST 2/4: CAD, FABRICATION AND EMBEDDED

When	Environment	Software	Status	Purpose / decision rule	Link
M5	macOS	FreeCAD - Apple silicon	Required path	Parametric CAD, assemblies/constraints and technical drawings.	Open
M5	macOS	Autodesk Fusion for personal use	Alternative	Use instead of FreeCAD if licensing and workflow suit the project.	Open
M5	macOS	Crealty Print	Required	Slicer and printer workflow for the owned CR-10 SE.	Open
M6	macOS	KiCad	Required	Schematics, connector maps and simple PCBs.	Open
M6	VS Code / macOS	PlatformIO IDE	Required first MCU path	ESP32 project setup, libraries, serial monitor and debugging.	Open
M6	macOS	ESP-IDF	Required later	Native ESP32-S3 framework after introductory PlatformIO labs.	Open
M6-M7	macOS	STM32CubeIDE	Optional	Install only if the optional STM32 Nucleo path is selected.	Open
M6	macOS	Saleae Logic 2 or PulseView	Required with analyzer	Protocol decoding; use the application matching the purchased hardware.	Open

Installation policy

Install only the software needed for the current module. Record exact versions and setup notes in the repository. Prefer official installers and documentation; do not mix ROS or Gazebo distributions inside one workspace.

SOFTWARE INSTALLATION CHECKLIST 3/4: ROS 2 AND SIMULATION

When	Environment	Software	Status	Purpose / decision rule	Link
M12	Ubuntu 24.04 VM	ROS 2 Jazzy Desktop	Required	Supported long-lived ROS baseline for Ubuntu Noble ARM64/x86_64.	Open
M12	Ubuntu 24.04 VM	colcon, rosdep and vcstool	Required	Workspace build, dependency resolution and multi-repository import.	Open
M12	Ubuntu 24.04 VM	RViz 2, rqt and rosbag2	Required	Visualization, introspection, plotting and reproducible data replay.	Open
M12	Ubuntu 24.04 VM	ros2_control + controllers	Required	Common simulated and physical joint/hardware abstraction.	Open
M13	Ubuntu 24.04 VM	Gazebo Harmonic	Required	Primary rigid-body, sensor and contact simulator.	Open
M13	Ubuntu 24.04 VM	gz_ros2_control	Required	Connect Gazebo joints to the ROS 2 control architecture.	Open
M12-M13	Ubuntu / macOS	Foxglove	Recommended	Inspect live/recorded ROS telemetry and share dashboards.	Open
M13	Ubuntu 24.04 VM	PlotJuggler	Recommended	High-rate control, IMU, current and latency plots.	Open

Installation policy

Install only the software needed for the current module. Record exact versions and setup notes in the repository. Prefer official installers and documentation; do not mix ROS or Gazebo distributions inside one workspace.

SOFTWARE INSTALLATION CHECKLIST 4/4: PERCEPTION, AUTONOMY AND AI

When	Environment	Software	Status	Purpose / decision rule	Link
M14	Ubuntu VM	OpenCV + Open3D	Required	Calibration, image processing, RGB-D and point-cloud labs.	Open
M15	Ubuntu VM	slam_toolbox	Required breadth	2D SLAM and mapping evidence; LiDAR purchase remains optional.	Open
M16	Ubuntu VM	Nav2	Breadth lab	Navigation breadth evidence, not the legged capstone controller.	Open
M17	Ubuntu VM	Movelt 2	Breadth lab	Manipulation planning breadth evidence, not the final project.	Open
M18	Ubuntu / macOS	JupyterLab + PyTorch	Required	ML experiments, evaluation, export and reproducible notebooks.	Open
M20-M21	Ubuntu VM	MuJoCo	Optional comparison	Second simulator for model/contact and learning comparisons.	Open
M20-M21	NVIDIA Linux or cloud	NVIDIA Isaac Lab	Conditional	Do not install on the M4; use only after a measured CUDA/RL need.	Open
M8-M20	Ubuntu VM	GTSAM	Optional advanced	Factor-graph estimation after the EKF/state-estimation baseline.	Open

Installation policy

Install only the software needed for the current module. Record exact versions and setup notes in the repository. Prefer official installers and documentation; do not mix ROS or Gazebo distributions inside one workspace.

PURCHASE PLAN A: OWNED EQUIPMENT AND FOUNDATION TOOLS

Gate	Qty	Priority	Item / specification	Decision rule	Link
Owned	1	Owned	MacBook Pro M4	Development, simulation and reports; no purchase.	-
Owned	1	Owned	Creality CR-10 SE	Printed prototypes and fixtures; no purchase.	-
M4	1	Required	Digital multimeter: continuity, diode mode and fused current input	Required before powered labs.	Open
M5	1	Required	150 mm metal caliper	Fits, shafts, bearings and print validation.	Open
M4	1 set	Required	Breadboards/protoboards plus R/C/LED/diode/BJT/MOSFET kit	Introductory electronics and permanent bench circuits.	Open
M4-M6	1 set	Required	Stripper, ratcheting crimper, ferrules, JST/Dupont, heat-shrink, wire and fuses	Safe repeatable wiring and harness repair.	Open
M5	Assort.	Required	M2/M3/M4 fasteners, washers, locknuts and heat-set inserts	Revised mechanical prototypes.	Open
M5	2+1+1 kg	Required	PLA + PETG + small TPU spool	Learning/fixtures, structural prototypes and compliant feet.	Open

Availability policy

Links were opened or found as live pages on 31 August 2026. Retail stock and price can change. Recheck the page immediately before purchase and select by measured requirements, documentation, voltage, interface and safety - never by the link alone.

PURCHASE PLAN B: ELECTRONICS AND ACTUATOR BENCH

Gate	Qty	Priority	Item / specification	Decision rule	Link
M4	1	Required	Temperature-controlled soldering station plus solder, flux and tip cleaner	Reliable joints; hot air is optional.	Open
M4-M7	1	Required	Fused CC/CV bench PSU, about 0-30 V and 5-10 A	Current-limited commissioning.	Open
M6	1	Required	8-channel logic analyzer	UART/I2C/SPI timing and decoding; genuine Saleae is optional.	Open
M10	1	Optional	Two-channel or better oscilloscope	Buy when PWM/noise/control diagnostics exceed analyzer capability.	Open
M6	2	Required	ESP32-S3-DevKitC-1	Primary beginner MCU plus a spare/second telemetry node.	Open
M6-M7	1	Optional	NUCLEO-G474RE	Motor-control/real-time specialization after ESP32.	Open
M7	1	Required	12 V geared DC motor with quadrature encoder	Bench identification; not the final leg actuator.	Open
M7	1	Required	Motor driver sized from measured motor current	Protected bench motor control; example is not a universal choice.	Open

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PURCHASE PLAN C: SENSING, COMPUTE AND MECHANICAL INTEGRATION

Gate	Qty	Priority	Item / specification	Decision rule	Link
M6-M11	1+1	Required	BNO086 IMU breakout or documented raw-IMU alternative	Orientation baseline; retain a spare for the final robot.	Open
M7	1+	Required	INA260 or correctly rated current sensor	Bench logging; final count/range follows the power architecture.	Open
M5-M23	As designed	Required	Bearings, shafts, spacers, couplings, thread-lock, inserts and compliant feet	Buy from released CAD/BOM dimensions.	Open
M14	1	Required	Orbbec Gemini 2 or supported RGB-D camera	Depth, point-cloud and terrain labs.	Open
M15	1	Optional	RPLIDAR A1	Dedicated 2D SLAM lab; depth-only evidence is acceptable.	Open
M13-M22	1	Required later	Raspberry Pi 5 (8 GB) or small Ubuntu-capable x86 PC	ROS 2 and classical onboard perception/control.	Open
M18+	1	Conditional	Jetson developer kit	Buy only if profiling proves a CUDA/perception/RL need.	Open
M22	1 set	Required	Support/tether rig, IR thermometer and spare harness parts	Safe commissioning and thermal validation.	Open

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PURCHASE PLAN D: FINAL ACTUATOR AND POWER DECISION GATES

Gate	Qty	Priority	Item / specification	Decision rule	Link
Gate 1	1	Required	One candidate smart actuator plus interface	Characterize current, bus, backlash, output loading and temperature.	Open
Gate 1 alt.	1	Alternative	ROBOTIS XH430-class current-controllable actuator	Use when measured load/accuracy needs justify the cost.	Open
Advanced alt.	1	Conditional	Integrated BLDC actuator such as CubeMars AK70-10	Only after high-voltage, CAN and power-system review.	Open
Gate 2	2 more	Required after pass	Matching actuators for one complete 3-DOF leg	Scale only after one-actuator thermal and bus tests.	Open
M22	9 more + 2 spares	Required after pass	Matching actuators for the 12-DOF quadruped	Buy only after the 3-DOF leg is calibrated and stable.	Open
M23	6 more + 2 spares	Required after pass	Matching actuators for the 18-DOF hexapod	Buy only after quadruped stand/gait/safety acceptance.	Open
M22-M23	1 system	Required	Battery, matched charger, BMS, contactor/E-stop, branch fuses and DC-DC	Size from measured voltage/current/energy and fault budget.	Open
M22-M23	4 then 6	Recommended	Contact-capable feet or load/contact sensors	Improves contact confidence, load balance and slip evaluation.	Open

Availability policy

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STAGED PURCHASING GATES AND COST CONTROL

Stage	Allowed purchase	Evidence required before scaling
Foundation	Tools, components, MCU, caliper and print materials	Safe measurements, breadboard labs and reviewed wiring.
One actuator	One motor/servo, interface, driver/sensor and guarded fixture	Current/thermal/bus/encoder data with physical E-stop.
One leg	Two additional identical actuators and leg hardware	One-actuator limits and selected family frozen.
Quadruped	Nine additional actuators plus spares and final power tree	3-DOF leg calibration and closed-loop tests pass.
Hexapod	Six additional actuators plus spares and stronger distribution	Quadruped stand/gait/safety acceptance passes.
Jetson/CUDA	Jetson only after profiling	CPU/x86 path misses a measured perception/learning target.

Actuator warning

Catalogue stall torque is not a safe design value. Use estimated continuous torque, duty cycle, thermal behavior, gearbox backlash, output bearing load, bus bandwidth, replacement availability and total robot mass. The inexpensive ST3215 path suits a light slow demonstrator; it is not equivalent to a torque-controlled BLDC leg.

24-MODULE CURRICULUM MAP 1/2

Block	Topic	Verified artifact	Hours
M1	Engineering workflow: Python, modern C++ and Rust literacy	telemetry-core v1.0 with replay, tests and timing report	75
M2	Applied mathematics I: linear algebra, calculus and ODEs	robot-math notebook set with numerical and geometric tests	75
M3	Probability, statistics and numerical optimization	uncertainty-and-optimization lab pack	75
M4	Electronics and signal processing from zero	electronics lab notebook plus sampled-signal filter report	75
M5	CAD, mechanics, materials and manufacturing	parametric 3-DOF leg CAD, drawings, BOM and test coupons	75
M6	Embedded systems, buses, real-time behavior and safety	ESP32-S3 controller with IMU/encoder telemetry and fault tests	75
M7	Actuators, motor control and system identification	one-actuator bench with torque-speed-current-thermal model	75
M8	Robot geometry and kinematics	tested planar arm and 3-DOF leg kinematics library	75
M9	Rigid-body dynamics and physical modelling	identified joint plus simulated 2-link/leg dynamics	75
M10	Classical, digital and frequency-domain control	joint and body-attitude controller benchmark	75
M11	State-space control, estimation and constrained optimization	LQR/EKF/QP comparison on a controlled model	75
M12	ROS 2 production software and observability	multi-node ROS 2 telemetry and control stack v1.0	75

Depth rule

Every module ends in code, CAD, configuration, data, a benchmark or a failure test. Reading and video completion alone do not satisfy a module.

24-MODULE CURRICULUM MAP 2/2

Block	Topic	Verified artifact	Hours
M13	Simulation, digital twins and HIL	Gazebo digital twin with fidelity and HIL reports	75
M14	Computer vision and 3D perception	calibrated RGB-D terrain/perception evidence pack	75
M15	Localization, mapping and SLAM	2D/visual SLAM comparison with failure analysis	75
M16	Motion planning, decision making and autonomy	planning benchmark plus behavior-tree autonomy lab	75
M17	Manipulation, HRI, ethics, security and industrial breadth	small manipulator lab plus responsible-robotics threat/safety review	75
M18	Machine-learning foundations for robotics	reproducible robotics ML baseline with deployment profile	75
M19	Legged robotics: estimation, gait and footholds	contact-aided estimator and crawl-gait simulation	75
M20	Advanced legged control and whole-body methods	QP stance-force allocator with classical fallback	75
M21	Robot learning: imitation, RL and sim-to-real	IL/RL policy evaluation with OOD and fallback report	75
M22	Physical quadruped integration and calibration	physical quadruped release candidate	75
M23	Hexapod scaling, terrain and reliability	slow actively leveled 18-DOF hexapod v1.0	75
M24	Validation, systems engineering and professional release	public v1.0 release, technical report and design defense	75

Depth rule

Every module ends in code, CAD, configuration, data, a benchmark or a failure test. Reading and video completion alone do not satisfy a module.

DEPENDENCY PATH AND GATE LOGIC

Phase	Blocks	Dependency outcome
Foundation	M1-M6 + G1	Programming, math, statistics, electronics, signals, CAD, embedded safety.
Robot core	M7-M12 + G2	Actuation, kinematics, dynamics, control, estimation/optimization and ROS 2.
Autonomy and AI	M13-M18 + G3	Simulation, perception, SLAM, planning, manipulation breadth and ML foundations.
Legged systems	M19-M24 + G4	Gait/estimation, whole-body methods, learning, quadruped, hexapod and release.

HARD PREREQUISITES

- No motor power before M4 safety practice and M6 safe-state firmware.
- No complete actuator purchase before M7 characterization and one-leg evidence.
- No RL/IL before M18 ML foundations and M19 classical gait/estimator baseline.
- No hexapod scale-up before the physical quadruped passes M22 acceptance.
- No claim survives without M24 traceability, raw trial summaries and limitations.

M1. ENGINEERING WORKFLOW: PYTHON, MODERN C++ AND RUST LITERACY

MODULE GOAL

Build a reproducible Linux workspace; use Python and C++ as the primary robotics languages and gain practical Rust literacy for safe systems components.

VERIFIED ARTIFACT

telemetry-core v1.0 with replay, tests and timing report

Week	Focus	Practical anchor
1	Python and Linux fundamentals	Shell/files/processes; virtual environments and packages
2	Modern C++ fundamentals	Compilation/linking, values, references and const
3	Testing, debugging and Rust literacy	pytest and GoogleTest; fixtures and parameterization
4	Concurrency and telemetry	Threads, mutexes, bounded queues and races

ACCEPTANCE CRITERIA

- Rebuild on a clean Ubuntu environment
- Pass malformed-input and concurrency tests
- Explain C++ RAII and Rust ownership trade-offs with measured evidence

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M1. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. PYTHON AND LINUX FUNDAMENTALS

- Shell/files/processes; virtual environments and packages
- Python types, functions, classes, NumPy and plotting
- Git commits, branches, issues and experiment log
- Parse and visualize recorded sensor packets

WEEK 2. MODERN C++ FUNDAMENTALS

- Compilation/linking, values, references and const
- RAII, ownership, STL containers and algorithms
- CMake targets, warnings and formatting
- Implement the same packet parser in C++

WEEK 3. TESTING, DEBUGGING AND RUST LITERACY

- pytest and GoogleTest; fixtures and parameterization
- GDB/LLDB, sanitizers and core dumps
- Rust ownership, borrowing, Result and Cargo tests
- Port the packet parser or one CLI tool to Rust

WEEK 4. CONCURRENCY AND TELEMETRY

- Threads, mutexes, bounded queues and races
- Monotonic clocks, timestamps and latency percentiles
- Recorder/replayer and deterministic inputs
- Benchmark Python/C++/Rust process boundaries

Evidence package

telemetry-core v1.0 with replay, tests and timing report. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M1. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	Python Tutorial	Language fundamentals and standard library.	Open
Reference	C++ reference	Modern C++ language and library lookup.	Open
Book	The Rust Programming Language	Ownership, borrowing, errors, Cargo and concurrency.	Open
Book	Pro Git	Branches, collaboration and reproducible history.	Open
Tutorial	CMake Tutorial	Target-based builds and testing.	Open
Tutorial	GoogleTest Primer	Unit tests, fixtures and assertions in C++.	Open

PRACTICE LADDER

- Problem 1 - Python and Linux fundamentals: Parse and visualize recorded sensor packets. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Modern C++ fundamentals: Implement the same packet parser in C++. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Testing, debugging and Rust literacy: Port the packet parser or one CLI tool to Rust. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Concurrency and telemetry: Benchmark Python/C++/Rust process boundaries. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M2. APPLIED MATHEMATICS I: LINEAR ALGEBRA, CALCULUS AND ODES

MODULE GOAL

Build the mathematical language needed for geometry, dynamics, estimation and control from first principles.

VERIFIED ARTIFACT

robot-math notebook set with numerical and geometric tests

Week	Focus	Practical anchor
1	Vectors, matrices and units	Vectors, bases, dot/cross products and projections
2	Linear systems and least squares	Linear systems, Gaussian elimination and QR awareness
3	Calculus for motion	Derivatives as velocity/sensitivity and gradients
4	ODEs and numerical integration	First/second-order ODEs and initial conditions

ACCEPTANCE CRITERIA

- Solve diagnostic exercises without copying
- Compare analytic and numerical results
- State units, assumptions and numerical error

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M2. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. VECTORS, MATRICES AND UNITS

- Vectors, bases, dot/cross products and projections
- Matrix multiplication, transpose and inverse
- Rank, condition number and unit consistency
- Solve forces and coordinate-change exercises

WEEK 2. LINEAR SYSTEMS AND LEAST SQUARES

- Linear systems, Gaussian elimination and QR awareness
- Least squares, residuals and overdetermined models
- Eigenvalues/eigenvectors and physical interpretation
- Fit a line and actuator calibration model

WEEK 3. CALCULUS FOR MOTION

- Derivatives as velocity/sensitivity and gradients
- Integrals as accumulation, work and impulse
- Partial derivatives and Jacobian preview
- Differentiate noisy versus analytic trajectories

WEEK 4. ODES AND NUMERICAL INTEGRATION

- First/second-order ODEs and initial conditions
- Euler, semi-implicit Euler and Runge-Kutta
- Stability, time step and truncation error
- Simulate mass-spring-damper and pendulum

Evidence package

robot-math notebook set with numerical and geometric tests. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M2. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Course	MIT 18.06 Linear Algebra	Vectors, matrices, eigenvalues and least squares.	Open
Course	3Blue1Brown Linear Algebra	Geometric intuition for transformations and bases.	Open
Book	OpenStax Calculus Volume 1	Derivatives, integrals and worked exercises.	Open
Course	MIT Differential Equations	ODE modelling and solution methods.	Open
Tutorial	SciPy integration	Numerical integration and ODE solvers.	Open
Text	Numerical Linear Algebra notes	Conditioning, QR and stable numerical methods.	Open

PRACTICE LADDER

- Problem 1 - Vectors, matrices and units: Solve forces and coordinate-change exercises. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Linear systems and least squares: Fit a line and actuator calibration model. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Calculus for motion: Differentiate noisy versus analytic trajectories. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - ODEs and numerical integration: Simulate mass-spring-damper and pendulum. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M3. PROBABILITY, STATISTICS AND NUMERICAL OPTIMIZATION

MODULE GOAL

Understand uncertainty, data quality and optimization before using filters, machine learning or QP controllers.

VERIFIED ARTIFACT

uncertainty-and-optimization lab pack

Week	Focus	Practical anchor
1	Probability foundations	Random variables, distributions and expectation
2	Statistics and experiments	Sampling, confidence intervals and effect size
3	Numerical optimization	Objective, constraints, gradient and Hessian intuition
4	Conditioning and uncertainty	SVD, pseudoinverse and ill-conditioned systems

ACCEPTANCE CRITERIA

- Estimate uncertainty from measured data
- Detect an ill-conditioned solve
- Defend sample size and metric selection

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M3. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. PROBABILITY FOUNDATIONS

- Random variables, distributions and expectation
- Variance, covariance, correlation and independence
- Conditional probability and Bayes rule
- Model encoder and IMU noise samples

WEEK 2. STATISTICS AND EXPERIMENTS

- Sampling, confidence intervals and effect size
- Outliers, robust summaries and hypothesis awareness
- Train/validation/test leakage and repeatability
- Design a 30-trial measurement experiment

WEEK 3. NUMERICAL OPTIMIZATION

- Objective, constraints, gradient and Hessian intuition
- Gradient descent, line search and local minima
- Least-squares optimization and regularization
- Solve calibration with `scipy.optimize`

WEEK 4. CONDITIONING AND UNCERTAINTY

- SVD, pseudoinverse and ill-conditioned systems
- Covariance propagation by Jacobians
- Monte Carlo validation and bootstrap awareness
- Report uncertainty instead of a single number

Evidence package

uncertainty-and-optimization lab pack. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M3. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Think Stats, 2nd edition	Probability and statistics through computation.	Open
Reference	NIST Statistics Handbook	Experimental design, distributions and uncertainty.	Open
Course	Stanford Convex Optimization	Optimization theory, examples and exercises.	Open
Tutorial	SciPy optimize	Numerical minimization and least squares.	Open
Tutorial	CVXPY examples	Constrained optimization practice.	Open
Reference	NumPy linear algebra	SVD, least squares and conditioning tools.	Open

PRACTICE LADDER

- Problem 1 - Probability foundations: Model encoder and IMU noise samples. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Statistics and experiments: Design a 30-trial measurement experiment. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Numerical optimization: Solve calibration with `scipy.optimize`. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Conditioning and uncertainty: Report uncertainty instead of a single number. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M4. ELECTRONICS AND SIGNAL PROCESSING FROM ZERO

MODULE GOAL

Learn safe low-voltage electronics, measurement and the signal concepts required for sensors and control.

VERIFIED ARTIFACT

electronics lab notebook plus sampled-signal filter report

Week	Focus	Practical anchor
1	DC circuits and measurement	Voltage, current, resistance, power and energy
2	Components and switching	Capacitors, diodes, BJT/MOSFET and flyback
3	Sampling and filtering	Sampling rate, Nyquist, aliasing and quantization
4	Frequency and latency	Frequency response, bandwidth and phase lag

ACCEPTANCE CRITERIA

- Wire and fuse a reviewed low-voltage circuit
- Identify aliasing and filter delay
- No battery/high-current work before Gate 1

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M4. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. DC CIRCUITS AND MEASUREMENT

- Voltage, current, resistance, power and energy
- Ohm and Kirchhoff laws; series/parallel circuits
- Multimeter modes and safe probe placement
- Breadboard LED and divider with current limiting

WEEK 2. COMPONENTS AND SWITCHING

- Capacitors, diodes, BJT/MOSFET and flyback
- Pull-ups, logic levels, grounding and decoupling
- Wire gauge, connectors, crimping and fuses
- Switch a protected inductive load

WEEK 3. SAMPLING AND FILTERING

- Sampling rate, Nyquist, aliasing and quantization
- Low/high-pass and notch-filter intuition
- Moving average, IIR and complementary filters
- Filter noisy encoder and IMU records

WEEK 4. FREQUENCY AND LATENCY

- Frequency response, bandwidth and phase lag
- FFT and vibration-spectrum interpretation
- ADC/PWM timing and sensor latency
- Measure filter trade-off: noise versus delay

Evidence package

electronics lab notebook plus sampled-signal filter report. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M4. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	All About Circuits textbook	Free DC/AC, semiconductor and circuit fundamentals.	Open
Tutorial	SparkFun Electronics Tutorials	Breadboards, multimeters, logic levels and components.	Open
Course	MIT Signals and Systems	Sampling, filtering and frequency response.	Open
Book	The Scientist and Engineer Guide to DSP	Practical DSP explanations and exercises.	Open
Simulator	Falstad Circuit Simulator	Interactive low-voltage circuit experiments.	Open
Guide	Adafruit multimeter guide	Safe voltage, resistance, continuity and current measurement.	Open

PRACTICE LADDER

- Problem 1 - DC circuits and measurement: Breadboard LED and divider with current limiting. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Components and switching: Switch a protected inductive load. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Sampling and filtering: Filter noisy encoder and IMU records. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Frequency and latency: Measure filter trade-off: noise versus delay. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M5. CAD, MECHANICS, MATERIALS AND MANUFACTURING

MODULE GOAL

Design printable robot structures from sketches through assemblies, load cases, FEA and measured prototypes.

VERIFIED ARTIFACT

parametric 3-DOF leg CAD, drawings, BOM and test coupons

Week	Focus	Practical anchor
1	CAD foundations	Sketches, constraints, dimensions and parameters
2	Machine elements and tolerances	Fasteners, inserts, bearings, shafts and couplings
3	Statics and strength	Free-body diagrams, equilibrium and load paths
4	FEA and print validation	Boundary conditions, mesh and convergence

ACCEPTANCE CRITERIA

- Fully constrained, revisioned CAD
- Hand calculation agrees with FEA/test trend
- Document fit and failure observations

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M5. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. CAD FOUNDATIONS

- Sketches, constraints, dimensions and parameters
- Extrude, revolve, patterns, holes and fillets
- Components, joints, assemblies and interference
- Model a link, bracket and joint stack

WEEK 2. MACHINE ELEMENTS AND TOLERANCES

- Fasteners, inserts, bearings, shafts and couplings
- Fits, datums, tolerance stacks and basic GD&T
- Cable paths, service access and maintainability
- Create drawing and exploded assembly

WEEK 3. STATICS AND STRENGTH

- Free-body diagrams, equilibrium and load paths
- Stress, strain, shear and beam deflection
- Factor of safety and uncertain loads
- Hand-check a worst-case leg load

WEEK 4. FEA AND PRINT VALIDATION

- Boundary conditions, mesh and convergence
- FDM anisotropy, orientation and layer adhesion
- PLA/PETG/TPU material trade-offs
- Print and load-test a single-leg structure

Evidence package

parametric 3-DOF leg CAD, drawings, BOM and test coupons. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M5. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Manual	FreeCAD documentation	Sketches, constraints, Part Design and drawings.	Open
Tutorial	Autodesk Fusion learning	Parametric CAD and assemblies alternative.	Open
Book	Shigley Mechanical Engineering Design	Machine elements, fatigue and design calculations.	Open
Course	MIT Mechanics of Materials	Stress, strain, bending and deflection.	Open
Guide	SimScale FEA fundamentals	Boundary conditions, mesh and interpretation.	Open
Guide	Prusa material table	FDM material and print-orientation trade-offs.	Open

PRACTICE LADDER

- Problem 1 - CAD foundations: Model a link, bracket and joint stack. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Machine elements and tolerances: Create drawing and exploded assembly. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Statics and strength: Hand-check a worst-case leg load. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - FEA and print validation: Print and load-test a single-leg structure. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M6. EMBEDDED SYSTEMS, BUSES, REAL-TIME BEHAVIOR AND SAFETY

MODULE GOAL

Build safe MCU firmware with deterministic I/O, communications, watchdogs and an independently testable safe state.

VERIFIED ARTIFACT

ESP32-S3 controller with IMU/encoder telemetry and fault tests

Week	Focus	Practical anchor
1	MCU fundamentals	GPIO, PWM, ADC, timers and interrupts
2	Communication buses	UART, I2C, SPI and framing/checksums
3	Scheduling and real time	Periodic tasks, jitter, deadlines and priorities
4	Safety architecture	Hazard analysis, limits and fault state machine

ACCEPTANCE CRITERIA

- No unsafe restart after watchdog or E-stop
- Measured loop timing and bus traces
- Gate 1 review: schematic, code and fault evidence

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M6. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. MCU FUNDAMENTALS

- GPIO, PWM, ADC, timers and interrupts
- Memory, stack/heap and volatile awareness
- Datasheets, pin maps and logic-level checks
- Read buttons and command a protected output

WEEK 2. COMMUNICATION BUSES

- UART, I2C, SPI and framing/checksums
- CAN/RS-485 awareness, termination and arbitration
- Logic-analyzer protocol decoding
- Recover from disconnects and corrupt frames

WEEK 3. SCHEDULING AND REAL TIME

- Periodic tasks, jitter, deadlines and priorities
- FreeRTOS tasks, queues and synchronization
- Watchdog, brownout and persistent configuration
- Measure loop latency and missed deadlines

WEEK 4. SAFETY ARCHITECTURE

- Hazard analysis, limits and fault state machine
- Physical E-stop and power removal path
- Current, temperature and communication limits
- Independently verify safe state

Evidence package

ESP32-S3 controller with IMU/encoder telemetry and fault tests. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M6. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	ESP-IDF Programming Guide	ESP32-S3 peripherals, build and debugging.	Open
Book	Mastering the FreeRTOS Kernel	Tasks, scheduling, queues and synchronization.	Open
Guide	Memfault Embedded Systems	Watchdogs, faults, metrics and production debugging.	Open
Reference	CAN bus introduction	Frames, arbitration, termination and errors.	Open
Tutorial	Saleae Logic 2 guides	Capture and decode real digital buses.	Open
Handbook	NASA Systems Engineering Handbook	Safety, assurance, verification and engineering practice.	Open

PRACTICE LADDER

- Problem 1 - MCU fundamentals: Read buttons and command a protected output. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Communication buses: Recover from disconnects and corrupt frames. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Scheduling and real time: Measure loop latency and missed deadlines. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Safety architecture: Independently verify safe state. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M7. ACTUATORS, MOTOR CONTROL AND SYSTEM IDENTIFICATION

MODULE GOAL

Characterize one actuator before choosing or purchasing a complete robot actuator set.

VERIFIED ARTIFACT

one-actuator bench with torque-speed-current-thermal model

Week	Focus	Practical anchor
1	Actuator selection	DC, BLDC, smart servo and gearbox trade-offs
2	Bench instrumentation	Driver sizing, current sensing and fusing
3	Closed-loop joint control	Position/velocity/current loop hierarchy
4	System identification	Friction, deadband, delay and backlash

ACCEPTANCE CRITERIA

- Measured rather than catalogue-only selection
- Current and temperature stay inside limits
- Do not buy 12/18 actuators before this gate

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M7. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. ACTUATOR SELECTION

- DC, BLDC, smart servo and gearbox trade-offs
- Torque-speed curves, reflected inertia and efficiency
- Encoder resolution, backlash and compliance
- Create requirements and comparison matrix

WEEK 2. BENCH INSTRUMENTATION

- Driver sizing, current sensing and fusing
- Quadrature decoding and velocity estimation
- Thermal measurement and duty cycle
- Build guarded one-actuator fixture

WEEK 3. CLOSED-LOOP JOINT CONTROL

- Position/velocity/current loop hierarchy
- PID, feedforward, saturation and anti-windup
- Trajectory limits and soft/hard stops
- Tune a repeatable position loop

WEEK 4. SYSTEM IDENTIFICATION

- Friction, deadband, delay and backlash
- Step/chirp tests and parameter fitting
- Validation data separate from fit data
- Publish model limits and thermal envelope

Evidence package

one-actuator bench with torque-speed-current-thermal model. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M7. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Course	Control Tutorials: DC Motor	Motor modelling, speed and position control.	Open
Guide	Pololu motor selection	Motor, gearbox and driver selection principles.	Open
Tutorial	SimpleFOC documentation	Practical closed-loop motor-control concepts.	Open
Reference	ODrive documentation	BLDC control architecture, calibration and limits.	Open
Course	MIT Underactuated system identification	Model fitting and validation in dynamic systems.	Open
Guide	DYNAMIXEL e-Manual	Smart-actuator modes, buses, limits and telemetry.	Open

PRACTICE LADDER

- Problem 1 - Actuator selection: Create requirements and comparison matrix. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Bench instrumentation: Build guarded one-actuator fixture. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Closed-loop joint control: Tune a repeatable position loop. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - System identification: Publish model limits and thermal envelope. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M8. ROBOT GEOMETRY AND KINEMATICS

MODULE GOAL

Implement coordinate frames, rotations, FK, IK and Jacobians for manipulators and legged robots.

VERIFIED ARTIFACT

tested planar arm and 3-DOF leg kinematics library

Week	Focus	Practical anchor
1	Frames and rotations	Coordinate frames, homogeneous transforms and chains
2	Forward kinematics	Link/joint models and Denavit-Hartenberg awareness
3	Inverse kinematics	Analytic IK, branches and reachability
4	Jacobians and singularities	Velocity Jacobian and finite-difference check

ACCEPTANCE CRITERIA

- FK/IK round-trip within declared tolerance
- Jacobian passes numerical differentiation
- Unreachable/singular requests fail safely

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M8. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. FRAMES AND ROTATIONS

- Coordinate frames, homogeneous transforms and chains
- $SO(2)/SO(3)$, rotation matrices and quaternions
- Composition order, active/passive conventions
- Visual frame tests with known poses

WEEK 2. FORWARD KINEMATICS

- Link/joint models and Denavit-Hartenberg awareness
- Product-of-exponentials and transform trees
- Planar 2-link and 3-DOF leg FK
- Compare CAD, analytic and numerical poses

WEEK 3. INVERSE KINEMATICS

- Analytic IK, branches and reachability
- Numerical IK and residual minimization
- Joint limits, seeds and failure detection
- Map foot targets into joint commands

WEEK 4. JACOBIANS AND SINGULARITIES

- Velocity Jacobian and finite-difference check
- Force mapping by Jacobian transpose
- Workspace and singularity metrics
- Build unit and property tests

Evidence package

tested planar arm and 3-DOF leg kinematics library. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M8. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Modern Robotics	Frames, FK, IK, Jacobians and exercises.	Open
Course	Modern Robotics course materials	Video lectures, assignments and code.	Open
Book	Robotics, Vision and Control	Applied kinematics with executable examples.	Open
Library	Robotics Toolbox for Python	Reference implementations and exercises.	Open
Library	Pinocchio documentation	Rigid-body geometry and Jacobian APIs.	Open
Notes	Drake multibody documentation	Frames, bodies, joints and computation.	Open

PRACTICE LADDER

- Problem 1 - Frames and rotations: Visual frame tests with known poses. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Forward kinematics: Compare CAD, analytic and numerical poses. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Inverse kinematics: Map foot targets into joint commands. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Jacobians and singularities: Build unit and property tests. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M9. RIGID-BODY DYNAMICS AND PHYSICAL MODELLING

MODULE GOAL

Move from free-body diagrams to joint, leg and body dynamics with explicit assumptions and validation.

VERIFIED ARTIFACT

identified joint plus simulated 2-link/leg dynamics

Week	Focus	Practical anchor
1	Newton-Euler foundations	Mass, COM, inertia, force, moment and energy
2	Lagrangian modelling	Generalized coordinates and kinetic/potential energy
3	Contact and friction	Normal force, friction cone and stick/slip
4	Model validation	Use M7 identification in a joint model

ACCEPTANCE CRITERIA

- Dynamics dimensions and energy checks pass
- Simulation reproduces measured trend
- State where the model is invalid

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M9. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. NEWTON-EULER FOUNDATIONS

- Mass, COM, inertia, force, moment and energy
- Translational and rotational equations of motion
- Free-body diagrams for link and leg
- Check conservation and units

WEEK 2. LAGRANGIAN MODELLING

- Generalized coordinates and kinetic/potential energy
- Mass matrix, Coriolis and gravity structure
- Two-link manipulator dynamics
- Compare symbolic/numeric simulation

WEEK 3. CONTACT AND FRICTION

- Normal force, friction cone and stick/slip
- Impact and compliant-contact modelling
- Support polygon and static margin
- Test contact-model sensitivity

WEEK 4. MODEL VALIDATION

- Use M7 identification in a joint model
- Compare measured and simulated trajectories
- Parameter sensitivity and model mismatch
- Define a fidelity budget

Evidence package

identified joint plus simulated 2-link/leg dynamics. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M9. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Modern Robotics dynamics chapters	Manipulator equations and inverse dynamics.	Open
Course	MIT Underactuated Robotics	Dynamics, contact and physically meaningful models.	Open
Course	ETH Robot Dynamics	Rigid-body dynamics for legged and mobile robots.	Open
Reference	Drake multibody documentation	Simulation and dynamics modelling APIs.	Open
Reference	MuJoCo computation	Contact, constraints and numerical dynamics.	Open
Book	Rigid Body Dynamics Algorithms	Spatial-vector algorithms and implementation.	Open

PRACTICE LADDER

- Problem 1 - Newton-Euler foundations: Check conservation and units. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Lagrangian modelling: Compare symbolic/numeric simulation. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Contact and friction: Test contact-model sensitivity. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Model validation: Define a fidelity budget. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M10. CLASSICAL, DIGITAL AND FREQUENCY-DOMAIN CONTROL

MODULE GOAL

Develop stable feedback control from transfer functions and sampled systems through a measured attitude controller.

VERIFIED ARTIFACT

joint and body-attitude controller benchmark

Week	Focus	Practical anchor
1	Feedback foundations	Open/closed loop, poles, zeros and stability
2	Frequency-domain reasoning	Bode magnitude/phase and bandwidth
3	Digital implementation	Sampling, zero-order hold and discretization
4	Attitude and balance lab	Roll/pitch target control using IMU

ACCEPTANCE CRITERIA

- Stable under declared delay and saturation
- Measured margins or equivalent robustness evidence
- Controller failure enters safe state

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M10. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. FEEDBACK FOUNDATIONS

- Open/closed loop, poles, zeros and stability
- Step response, rise, overshoot and settling
- PID terms, feedforward and anti-windup
- Tune on simulation then bench

WEEK 2. FREQUENCY-DOMAIN REASONING

- Bode magnitude/phase and bandwidth
- Gain/phase margin and disturbance rejection
- Noise amplification and filter interaction
- Measure closed-loop frequency response

WEEK 3. DIGITAL IMPLEMENTATION

- Sampling, zero-order hold and discretization
- Delay, jitter, quantization and saturation
- Rate selection and multirate awareness
- Validate embedded loop timing

WEEK 4. ATTITUDE AND BALANCE LAB

- Roll/pitch target control using IMU
- Gain scheduling and limit handling
- Incline and repeatable disturbance tests
- Report RMSE, peak error and settling

Evidence package

joint and body-attitude controller benchmark. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M10. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Feedback Systems	Free control textbook with exercises.	Open
Course	Control Tutorials for MATLAB and Simulink	Classical and state-space worked problems.	Open
Library	Python Control Systems Library	Simulation, Bode, margins and design tools.	Open
Course	MIT Underactuated feedback control	Stability and nonlinear feedback intuition.	Open
Tutorial	PID Explained	Practical PID behavior and tuning intuition.	Open
Reference	SciPy signal	Discrete systems, filters and frequency response.	Open

PRACTICE LADDER

- Problem 1 - Feedback foundations: Tune on simulation then bench. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Frequency-domain reasoning: Measure closed-loop frequency response. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Digital implementation: Validate embedded loop timing. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Attitude and balance lab: Report RMSE, peak error and settling. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M11. STATE-SPACE CONTROL, ESTIMATION AND CONSTRAINED OPTIMIZATION

MODULE GOAL

Connect state-space models, observability, Kalman estimation, LQR and constrained optimization.

VERIFIED ARTIFACT

LQR/EKF/QP comparison on a controlled model

Week	Focus	Practical anchor
1	State-space systems	Continuous/discrete state-space models
2	Kalman estimation	Prediction/update, process and measurement noise
3	Optimal control	Quadratic costs and LQR
4	Constrained optimization	QP formulation, bounds and infeasibility

ACCEPTANCE CRITERIA

- Explain observable and unobservable states
- Innovation statistics are plausible
- Optimization failure is tested, not ignored

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M11. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. STATE-SPACE SYSTEMS

- Continuous/discrete state-space models
- Linearization around an operating point
- Controllability and observability
- Build state transition tests

WEEK 2. KALMAN ESTIMATION

- Prediction/update, process and measurement noise
- Covariance meaning, tuning and consistency
- EKF linearization and residual gating
- Estimate attitude from noisy sensors

WEEK 3. OPTIMAL CONTROL

- Quadratic costs and LQR
- LQG relationship and separation awareness
- Trade state error against control effort
- Compare PID and LQR fairly

WEEK 4. CONSTRAINED OPTIMIZATION

- QP formulation, bounds and infeasibility
- Friction/force allocation toy problem
- Warm starts and solve-time distribution
- Fallback when solver fails

Evidence package

LQR/EKF/QP comparison on a controlled model. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M11. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Kalman and Bayesian Filters in Python	Executable filtering derivations and exercises.	Open
Course	Stanford EE363 Linear Dynamical Systems	State space, controllability, estimation and LQR.	Open
Book	Convex Optimization	QP theory and exercises.	Open
Tutorial	CVXPY tutorials	Constrained controller and estimator problems.	Open
Solver	OSQP documentation	Real-time-friendly quadratic programming.	Open
Library	FilterPy	Kalman/EKF implementations for controlled experiments.	Open

PRACTICE LADDER

- Problem 1 - State-space systems: Build state transition tests. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Kalman estimation: Estimate attitude from noisy sensors. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Optimal control: Compare PID and LQR fairly. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Constrained optimization: Fallback when solver fails. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M12. ROS 2 PRODUCTION SOFTWARE AND OBSERVABILITY

MODULE GOAL

Build a production-style ROS 2 stack with interfaces, timing, lifecycle, diagnostics and deterministic replay.

VERIFIED ARTIFACT

multi-node ROS 2 telemetry and control stack v1.0

Week	Focus	Practical anchor
1	ROS graph and interfaces	Nodes, topics, services, actions and parameters
2	DDS and execution	RMW/DDS discovery and domains
3	Frames and hardware	TF2 timestamps and buffering
4	Diagnostics and replay	rosbag2, tracing, diagnostics and logging

ACCEPTANCE CRITERIA

- Clean build and launch from documentation
- QoS/timing choices supported by evidence
- Gate 2 architecture and replay review

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M12. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. ROS GRAPH AND INTERFACES

- Nodes, topics, services, actions and parameters
- Custom messages, IDL and serialization
- Launch, composition and namespaces
- Define interface ownership and units

WEEK 2. DDS AND EXECUTION

- RMW/DDS discovery and domains
- QoS reliability, durability, history and deadlines
- Executors, callback groups and intra-process
- Measure callback latency and drops

WEEK 3. FRAMES AND HARDWARE

- TF2 timestamps and buffering
- URDF/Xacro, inertials and collision geometry
- ros2_control hardware interfaces
- Lifecycle-managed hardware mock

WEEK 4. DIAGNOSTICS AND REPLAY

- rosbag2, tracing, diagnostics and logging
- PlotJuggler/Foxglove dashboards
- launch_testing and fault injection
- Reproduce a recorded failure

Evidence package

multi-node ROS 2 telemetry and control stack v1.0. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M12. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	ROS 2 Jazzy tutorials	Nodes, topics, services, actions and parameters.	Open
Guide	ROS 2 Concepts	Architecture and middleware concepts.	Open
Guide	ROS 2 QoS	Reliability, durability, history and deadlines.	Open
Tutorial	tf2 tutorials	Timestamped frame transforms and debugging.	Open
Tutorial	ros2_control getting started	Hardware interfaces and controllers.	Open
Guide	ROS 2 tracing	Measure callbacks, executors and latency.	Open

PRACTICE LADDER

- Problem 1 - ROS graph and interfaces: Define interface ownership and units. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - DDS and execution: Measure callback latency and drops. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Frames and hardware: Lifecycle-managed hardware mock. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Diagnostics and replay: Reproduce a recorded failure. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M13. SIMULATION, DIGITAL TWINS AND HIL

MODULE GOAL

Create a measured digital twin and use SIL/PIL/HIL to retire risk before multi-actuator hardware.

VERIFIED ARTIFACT

Gazebo digital twin with fidelity and HIL reports

Week	Focus	Practical anchor
1	Physics simulation	SDF/Gazebo worlds, contacts and friction
2	Robot digital twin	Quadruped URDF/Xacro and transmissions
3	SIL and PIL	Controller-in-loop and firmware-in-loop boundaries
4	HIL and fidelity	One-leg HIL with real communications

ACCEPTANCE CRITERIA

- No visual-only simulation claim
- Contact and timing failures reproduced
- Model gaps become tracked engineering work

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M13. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. PHYSICS SIMULATION

- SDF/Gazebo worlds, contacts and friction
- Joint limits, damping and actuator models
- Sensor noise, rate and delay
- Validate simple analytic cases

WEEK 2. ROBOT DIGITAL TWIN

- Quadruped URDF/Xacro and transmissions
- IMU, joints, contacts and depth sensor
- Ground truth and test fixtures
- Automate headless simulation tests

WEEK 3. SIL AND PIL

- Controller-in-loop and firmware-in-loop boundaries
- Deterministic seeds and scenario catalog
- Timing/fault injection and pass criteria
- Compare real-time factor and latency

WEEK 4. HIL AND FIDELITY

- One-leg HIL with real communications
- Sim-real parameter alignment
- Fidelity budget by subsystem
- Decide what must be tested physically

Evidence package

Gazebo digital twin with fidelity and HIL reports. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M13. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	Gazebo Harmonic tutorials	Worlds, models, sensors, transport and plugins.	Open
Guide	Gazebo physics parameters	Solvers, contacts, friction and time steps.	Open
Tutorial	gz_ros2_control	Joint-control integration with ROS 2.	Open
Reference	SDF specification	Robot/world model schema and semantics.	Open
Reference	MuJoCo documentation	Alternative physics and contact simulator.	Open
Guide	ROS 2 launch testing	Automated integration and simulation tests.	Open

PRACTICE LADDER

- Problem 1 - Physics simulation: Validate simple analytic cases. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Robot digital twin: Automate headless simulation tests. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - SIL and PIL: Compare real-time factor and latency. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - HIL and fidelity: Decide what must be tested physically. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M14. COMPUTER VISION AND 3D PERCEPTION

MODULE GOAL

Calibrate cameras and build classical/deep perception pipelines with geometric and statistical evaluation.

VERIFIED ARTIFACT

calibrated RGB-D terrain/perception evidence pack

Week	Focus	Practical anchor
1	Camera geometry	Pinhole model, distortion and calibration
2	Classical vision	Color/edges/features and geometric fitting
3	Point clouds and 3D	Depth-to-point-cloud and frame transforms
4	Deep perception	CNN features, transfer learning and augmentation

ACCEPTANCE CRITERIA

- Calibration and timestamps documented
- Dataset leakage checks pass
- Performance and latency both reported

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M14. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. CAMERA GEOMETRY

- Pinhole model, distortion and calibration
- Stereo/depth geometry and uncertainty
- Hand-eye/frame calibration awareness
- Measure reprojection error

WEEK 2. CLASSICAL VISION

- Color/edges/features and geometric fitting
- Optical flow and tracking awareness
- Robust estimation with RANSAC
- Replay deterministic image tests

WEEK 3. POINT CLOUDS AND 3D

- Depth-to-point-cloud and frame transforms
- Filtering, normals, registration and ICP
- Terrain plane and height representation
- Measure latency and geometric error

WEEK 4. DEEP PERCEPTION

- CNN features, transfer learning and augmentation
- Segmentation/detection baseline
- Dataset splits, precision/recall and calibration
- Compare deep and classical baselines

Evidence package

calibrated RGB-D terrain/perception evidence pack. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M14. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	OpenCV tutorials	Calibration, geometry, features and image processing.	Open
Tutorial	Open3D geometry tutorials	Point clouds, registration and visualization.	Open
Course	Stanford CS231n	Deep vision foundations and assignments.	Open
Book	Computer Vision: Algorithms and Applications	Free vision textbook and exercises.	Open
Dataset	TUM RGB-D dataset	Calibrated RGB-D sequences and ground truth.	Open
Tutorial	OpenCV camera calibration	Reprojection error and calibration practice.	Open

PRACTICE LADDER

- Problem 1 - Camera geometry: Measure reprojection error. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Classical vision: Replay deterministic image tests. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Point clouds and 3D: Measure latency and geometric error. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Deep perception: Compare deep and classical baselines. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M15. LOCALIZATION, MAPPING AND SLAM

MODULE GOAL

Implement and compare odometry, filtering, scan matching and graph-based SLAM using quantitative trajectory metrics.

VERIFIED ARTIFACT

2D/visual SLAM comparison with failure analysis

Week	Focus	Practical anchor
1	Odometry and localization	Dead reckoning and error growth
2	Mapping and scan matching	Occupancy grids and inverse sensor model
3	Graph SLAM	Pose graphs, factors and loop closure
4	System evaluation	ATE/RPE and map-quality metrics

ACCEPTANCE CRITERIA

- Quantitative trajectory comparison
- Failure cases and parameter sensitivity reported
- SLAM remains breadth for final legged robot

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M15. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. ODOMETRY AND LOCALIZATION

- Dead reckoning and error growth
- Wheel/leg/visual/inertial odometry concepts
- EKF localization and frame conventions
- Evaluate drift against ground truth

WEEK 2. MAPPING AND SCAN MATCHING

- Occupancy grids and inverse sensor model
- ICP/scan matching and degeneracy
- AMCL/particle-filter intuition
- Build 2D LiDAR replay lab

WEEK 3. GRAPH SLAM

- Pose graphs, factors and loop closure
- Nonlinear least squares and robust losses
- GTSAM awareness and covariance
- Inspect false loop closures

WEEK 4. SYSTEM EVALUATION

- ATE/RPE and map-quality metrics
- Time synchronization and calibration effects
- Kidnapped/dropout/degraded cases
- Compare LiDAR and visual pipelines

Evidence package

2D/visual SLAM comparison with failure analysis. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M15. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Probabilistic Robotics course materials	Localization, mapping and Bayesian robotics.	Open
Book	Factor Graphs for Robot Perception	Modern SLAM and factor-graph derivations.	Open
Library	GTSAM tutorials	Factor graphs and nonlinear estimation.	Open
Tutorial	slam_toolbox documentation	ROS 2 mapping, localization and lifelong modes.	Open
Tool	evo trajectory evaluation	ATE/RPE evaluation and plots.	Open
Dataset	KITTI odometry benchmark	Trajectory data and quantitative benchmarking.	Open

PRACTICE LADDER

- Problem 1 - Odometry and localization: Evaluate drift against ground truth. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Mapping and scan matching: Build 2D LiDAR replay lab. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Graph SLAM: Inspect false loop closures. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - System evaluation: Compare LiDAR and visual pipelines. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M16. MOTION PLANNING, DECISION MAKING AND AUTONOMY

MODULE GOAL

Cover common search, sampling, trajectory and task-planning methods before specializing in footholds.

VERIFIED ARTIFACT

planning benchmark plus behavior-tree autonomy lab

Week	Focus	Practical anchor
1	Discrete search	State graphs, BFS, Dijkstra and A*
2	Sampling-based planning	Configuration space and collision checking
3	Trajectory generation	Time scaling, splines and smoothness
4	Task autonomy	Finite-state machines and behavior trees

ACCEPTANCE CRITERIA

- Compare at least two planners
- Planning failure triggers recovery
- Explain search versus trajectory optimization

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M16. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. DISCRETE SEARCH

- State graphs, BFS, Dijkstra and A*
- Admissible heuristics and cost maps
- Replanning and dynamic obstacles
- Benchmark expansions, cost and runtime

WEEK 2. SAMPLING-BASED PLANNING

- Configuration space and collision checking
- PRM, RRT and RRT* concepts
- Completeness and narrow passages
- MoveIt 2 planning-scene lab

WEEK 3. TRAJECTORY GENERATION

- Time scaling, splines and smoothness
- Velocity/acceleration/jerk constraints
- Trajectory optimization awareness
- Reject infeasible commands

WEEK 4. TASK AUTONOMY

- Finite-state machines and behavior trees
- Recovery, timeout and degraded modes
- MDP/POMDP conceptual foundation
- Nav2 breadth lab with measurable recovery

Evidence package

planning benchmark plus behavior-tree autonomy lab. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M16. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Planning Algorithms	Free search, sampling and decision-planning text.	Open
Library	OMPL documentation	Sampling-based planning algorithms and demos.	Open
Tutorial	Movelt 2 tutorials	Planning scenes, collision checking and trajectories.	Open
Tutorial	Nav2 Jazzy getting started	Navigation, costmaps, planners and recovery.	Open
Tutorial	BehaviorTree.CPP tutorials	Task logic, recovery and reactive execution.	Open
Course	MIT Underactuated planning	Trajectory optimization and planning through dynamics.	Open

PRACTICE LADDER

- Problem 1 - Discrete search: Benchmark expansions, cost and runtime. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Sampling-based planning: Movelt 2 planning-scene lab. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Trajectory generation: Reject infeasible commands. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Task autonomy: Nav2 breadth lab with measurable recovery. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M17. MANIPULATION, HRI, ETHICS, SECURITY AND INDUSTRIAL BREADTH

MODULE GOAL

Gain practical literacy in major robotics branches outside legged locomotion without pretending specialist depth.

VERIFIED ARTIFACT

small manipulator lab plus responsible-robotics threat/safety review

Week	Focus	Practical anchor
1	Manipulation fundamentals	Arm FK/IK, collision and grasp frames
2	Human-robot interaction	Teleoperation, autonomy levels and override
3	Ethics and cybersecurity	Privacy, dataset bias and responsible AI
4	Industrial and team context	PLCs, fieldbuses and industrial robot cells

ACCEPTANCE CRITERIA

- One reproducible manipulation lab
- Document ethical/security decisions
- Do not claim certification from awareness work

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M17. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. MANIPULATION FUNDAMENTALS

- Arm FK/IK, collision and grasp frames
- Pick/place planning in MoveIt 2
- Impedance/force-control concepts
- Evaluate repeatability and failure causes

WEEK 2. HUMAN-ROBOT INTERACTION

- Teleoperation, autonomy levels and override
- Operator feedback, usability and mode errors
- Shared workspace and safe speed awareness
- Run a small operator task study

WEEK 3. ETHICS AND CYBERSECURITY

- Privacy, dataset bias and responsible AI
- Threat model: network, update and command paths
- Authentication/encryption/update awareness
- Separate safety and security hazards

WEEK 4. INDUSTRIAL AND TEAM CONTEXT

- PLCs, fieldbuses and industrial robot cells
- ISO 10218/13849 and IEC 61508 awareness
- Requirements, reviews and change control
- Team/internship experience gap plan

Evidence package

small manipulator lab plus responsible-robotics threat/safety review. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M17. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Course	MIT Robotic Manipulation	Kinematics, grasping, planning and contact.	Open
Tutorial	Movelt pick and place	Reproducible manipulation planning lab.	Open
Guide	ROS-Industrial training	Industrial ROS workflows and exercises.	Open
Reference	NIST Cybersecurity Framework	Threat and risk management vocabulary.	Open
Reference	ISO 10218 overview	Industrial robot safety standard context.	Open
Book	Human-Robot Interaction	Open HRI textbook, methods and exercises.	Open

PRACTICE LADDER

- Problem 1 - Manipulation fundamentals: Evaluate repeatability and failure causes. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Human-robot interaction: Run a small operator task study. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Ethics and cybersecurity: Separate safety and security hazards. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Industrial and team context: Team/internship experience gap plan. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M18. MACHINE-LEARNING FOUNDATIONS FOR ROBOTICS

MODULE GOAL

Learn supervised learning, PyTorch, datasets and deployment measurement before imitation or reinforcement learning.

VERIFIED ARTIFACT

reproducible robotics ML baseline with deployment profile

Week	Focus	Practical anchor
1	Supervised learning	Regression/classification and loss functions
2	PyTorch workflow	Tensors, autograd, modules and optimizers
3	Evaluation and uncertainty	Confusion matrices, ROC/PR and calibration
4	Deployment	Export/quantization awareness

ACCEPTANCE CRITERIA

- Beat a simple baseline on held-out data
- Reproduce run from config and seed
- Gate 3 data/model/safety review

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M18. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. SUPERVISED LEARNING

- Regression/classification and loss functions
- Bias/variance, regularization and baselines
- Train/validation/test discipline
- Fit terrain or contact classifier

WEEK 2. PYTORCH WORKFLOW

- Tensors, autograd, modules and optimizers
- SGD/Adam, normalization and schedules
- Experiment configuration and seeds
- Track metrics and artifacts

WEEK 3. EVALUATION AND UNCERTAINTY

- Confusion matrices, ROC/PR and calibration
- Class imbalance and confidence thresholds
- OOD concepts and data shift
- Perform ablation and error taxonomy

WEEK 4. DEPLOYMENT

- Export/quantization awareness
- CPU/GPU latency, throughput and memory
- Sensor-to-decision timing
- Decide whether Jetson/CUDA is justified

Evidence package

reproducible robotics ML baseline with deployment profile. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M18. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	PyTorch tutorials	Tensors, autograd, models and deployment.	Open
Book	Dive into Deep Learning	Interactive deep-learning text and exercises.	Open
Course	Full Stack Deep Learning	Data, experiments, deployment and monitoring.	Open
Tutorial	scikit-learn model evaluation	Metrics, validation and leakage-aware workflows.	Open
Guide	PyTorch reproducibility	Seeds, nondeterminism and repeatable experiments.	Open
Guide	Google ML Rules	Practical baselines, metrics and production discipline.	Open

PRACTICE LADDER

- Problem 1 - Supervised learning: Fit terrain or contact classifier. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - PyTorch workflow: Track metrics and artifacts. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Evaluation and uncertainty: Perform ablation and error taxonomy. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Deployment: Decide whether Jetson/CUDA is justified. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M19. LEGGED ROBOTICS: ESTIMATION, GAIT AND FOOTHOLDS

MODULE GOAL

Combine body/leg frames, IMU, joint and contact evidence into safe slow quadruped locomotion.

VERIFIED ARTIFACT

contact-aided estimator and crawl-gait simulation

Week	Focus	Practical anchor
1	Contact-aided state estimation	IMU bias/noise and attitude propagation
2	Gait scheduling	Stance/swing state machine and leg transitions
3	Swing and footholds	Swing trajectories and foot clearance
4	Balance and recovery	Support margin and body leveling

ACCEPTANCE CRITERIA

- Stable simulated stand ≥ 60 s
- Contact/slip metrics and fall rate logged
- Classical gait works before learning

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M19. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. CONTACT-AIDED STATE ESTIMATION

- IMU bias/noise and attitude propagation
- Leg-odometry constraints and contact modes
- Contact confidence and slip residuals
- Dropout/bias/timing ablation

WEEK 2. GAIT SCHEDULING

- Stance/swing state machine and legal transitions
- Crawl/trot awareness and duty factor
- Startup, stop and phase changes
- Deterministic transition tests

WEEK 3. SWING AND FOOHOLDS

- Swing trajectories and foot clearance
- Reachability, collision and terrain height
- Conservative foothold score and fallback
- Log rejected foothold reasons

WEEK 4. BALANCE AND RECOVERY

- Support margin and body leveling
- Stance force distribution
- Repeatable small disturbances
- Degraded gait and safe stop

Evidence package

contact-aided estimator and crawl-gait simulation. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M19. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Course	ETH Robot Dynamics	Legged kinematics, dynamics, estimation and control context.	Open
Course	MIT Underactuated walking	Hybrid dynamics, contacts and locomotion.	Open
Paper	MIT Cheetah 3 locomotion control	Contact-aware robust locomotion architecture.	Open
Paper	ANYmal state estimation	Contact-aided state estimation for legged robots.	Open
Project	CHAMP quadruped controller	ROS-based gait and quadruped reference implementation.	Open
Guide	Unitree legged SDK	Real robot command/state and safety awareness.	Open

PRACTICE LADDER

- Problem 1 - Contact-aided state estimation: Dropout/bias/timing ablation. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Gait scheduling: Deterministic transition tests. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Swing and footholds: Log rejected foothold reasons. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Balance and recovery: Degraded gait and safe stop. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M20. ADVANCED LEGGED CONTROL AND WHOLE-BODY METHODS

MODULE GOAL

Implement a constrained classical stance controller and understand the path to whole-body control and MPC.

VERIFIED ARTIFACT

QP stance-force allocator with classical fallback

Week	Focus	Practical anchor
1	Centroidal modelling	COM, momentum and desired body wrench
2	Force optimization	Friction pyramids and normal-force limits
3	Whole-body architecture	Jacobian-transpose torque mapping
4	MPC and robustness	Finite-horizon prediction and constraints

ACCEPTANCE CRITERIA

- QP result respects contact/current constraints
- Fallback tested under solver failure
- No dynamic-running claim

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M20. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. CENTROIDAL MODELLING

- COM, momentum and desired body wrench
- Changing contact sets and force balance
- Model/unit/conservation checks
- Compare quasi-static and centroidal models

WEEK 2. FORCE OPTIMIZATION

- Friction pyramids and normal-force limits
- QP stance-force allocation
- Conditioning, infeasibility and warm starts
- Measure solver-time distribution

WEEK 3. WHOLE-BODY ARCHITECTURE

- Jacobian-transpose torque mapping
- Impedance and task priorities
- Joint/current limits and saturation
- Inverse-dynamics architecture awareness

WEEK 4. MPC AND ROBUSTNESS

- Finite-horizon prediction and constraints
- NMPC conceptual lab only
- Disturbance/incline benchmark
- Fallback to conservative stance

Evidence package

QP stance-force allocator with classical fallback. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M20. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Course	MIT Underactuated locomotion	Centroidal, contact and optimization concepts.	Open
Solver	OSQP documentation	QP formulation, warm starts and diagnostics.	Open
Library	Pinocchio documentation	Fast rigid-body algorithms and Jacobians.	Open
Library	Crocodyl documentation	Optimal control for articulated robots.	Open
Solver	acados documentation	Real-time optimal control and NMPC experiments.	Open
Paper	Whole-Body Dynamic Locomotion	Task/contact optimization architecture for legged robots.	Open

PRACTICE LADDER

- Problem 1 - Centroidal modelling: Compare quasi-static and centroidal models. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Force optimization: Measure solver-time distribution. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Whole-body architecture: Inverse-dynamics architecture awareness. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - MPC and robustness: Fallback to conservative stance. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M21. ROBOT LEARNING: IMITATION, RL AND SIM-TO-REAL

MODULE GOAL

Run one bounded learning experiment against a frozen classical baseline and independent safety monitor.

VERIFIED ARTIFACT

IL/RL policy evaluation with OOD and fallback report

Week	Focus	Practical anchor
1	Imitation learning	Demonstration collection and behavior cloning
2	Reinforcement learning	MDP, observation, action, reward and termination
3	Domain randomization	Mass, friction, delay, noise and terrain ranges
4	Safety and OOD	Confidence/OOD indicator and policy shield

ACCEPTANCE CRITERIA

- At least three seeds or justified alternative
- Frozen classical baseline remains available
- Learning is optional if safety gates fail

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M21. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. IMITATION LEARNING

- Demonstration collection and behavior cloning
- Distribution shift and compounding error
- DAgger awareness and dataset audit
- Compare with classical commands

WEEK 2. REINFORCEMENT LEARNING

- MDP, observation, action, reward and termination
- PPO or SAC - choose one
- Training curves, seeds and variance
- Avoid reward-only success claims

WEEK 3. DOMAIN RANDOMIZATION

- Mass, friction, delay, noise and terrain ranges
- Curriculum and randomization bounds
- Sim-to-real validation protocol
- Ablate each randomization family

WEEK 4. SAFETY AND OOD

- Confidence/OOD indicator and policy shield
- Command, current and posture limits
- Fallback to classical stand/gait pause
- Push-recovery comparison

Evidence package

IL/RL policy evaluation with OOD and fallback report. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M21. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Reinforcement Learning: An Introduction	Core RL theory and exercises.	Open
Course	OpenAI Spinning Up	Policy-gradient concepts and implementations.	Open
Library	Gymnasium documentation	Environment API, wrappers and reproducible evaluation.	Open
Library	Stable-Baselines3	PPO/SAC baselines and evaluation helpers.	Open
Tutorial	Isaac Lab locomotion environments	NVIDIA simulation/RL path, conditional on hardware.	Open
Paper	DAgger	Imitation-learning distribution shift and data aggregation.	Open

PRACTICE LADDER

- Problem 1 - Imitation learning: Compare with classical commands. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Reinforcement learning: Avoid reward-only success claims. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Domain randomization: Ablate each randomization family. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Safety and OOD: Push-recovery comparison. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M22. PHYSICAL QUADRUPEL INTEGRATION AND CALIBRATION

MODULE GOAL

Assemble and commission the 12-DOF robot in stages, aligning mechanics, power, buses, models and software.

VERIFIED ARTIFACT

physical quadruped release candidate

Week	Focus	Practical anchor
1	Mechanical and electrical assembly	Mass/COM and fastener/insert inspection
2	Actuator calibration	IDs, directions, zeros and joint limits
3	HIL and sim-to-real	Model parameter and latency alignment
4	Stand and slow gait	Stable stand and roll/pitch tracking

ACCEPTANCE CRITERIA

- Physical E-stop independently verified
- Stable stand ≥ 60 s
- No unbounded current, thermal or bus faults

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M22. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. MECHANICAL AND ELECTRICAL ASSEMBLY

- Mass/COM and fastener/insert inspection
- Harness, branch fusing and strain relief
- E-stop/contactors and support rig
- Continuity and power-up checklist

WEEK 2. ACTUATOR CALIBRATION

- IDs, directions, zeros and joint limits
- Encoder/current/temperature calibration
- Bus utilization and error monitoring
- One leg before four-leg enable

WEEK 3. HIL AND SIM-TO-REAL

- Model parameter and latency alignment
- Controller/estimator replay tests
- Contact/friction and IMU validation
- Track fidelity gaps

WEEK 4. STAND AND SLOW GAIT

- Stable stand and roll/pitch tracking
- Flat-floor crawl and controlled stop
- Slip/current/thermal limits
- Repeatable disturbance trials

Evidence package

physical quadruped release candidate. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M22. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Tutorial	ros2_control hardware components	Physical actuator/sensor interface architecture.	Open
Guide	ROS URDF tutorials	Robot descriptions, joints and validation.	Open
Guide	Foxglove ROS visualization	Build telemetry and debugging dashboards.	Open
Guide	Battery University lithium-ion safety	Battery selection and risk awareness.	Open
Guide	NASA fault tree handbook	Structured fault analysis and mitigations.	Open
Tutorial	pytest parametrization	Automate calibration and regression matrices.	Open

PRACTICE LADDER

- Problem 1 - Mechanical and electrical assembly: Continuity and power-up checklist. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Actuator calibration: One leg before four-leg enable. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - HIL and sim-to-real: Track fidelity gaps. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Stand and slow gait: Repeatable disturbance trials. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M23. HEXAPOD SCALING, TERRAIN AND RELIABILITY

MODULE GOAL

Scale only a proven architecture to 18 DOF and validate active leveling, six-contact estimation and degraded behavior.

VERIFIED ARTIFACT

slow actively leveled 18-DOF hexapod v1.0

Week	Focus	Practical anchor
1	Scale mechanics and power	Six hip frames, stiffness and service access
2	Six-leg gaits	Wave, ripple and conditional tripod gait
3	Terrain and contact	Six-contact estimator and terrain plane
4	Reliability and energy	Fault tree and injected node/leg faults

ACCEPTANCE CRITERIA

- Stable stand ≥ 120 s
- Wave/ripple gait and ± 10 -15 deg target subject to hardware
- 30-50+ trials or justified sample size

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M23. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. SCALE MECHANICS AND POWER

- Six hip frames, stiffness and service access
- 18-actuator current/thermal budget
- Branch fusing, BMS/DC-DC and bus load
- Buy six additional actuators only now

WEEK 2. SIX-LEG GAITS

- Wave, ripple and conditional tripod gait
- Support margins and legal transitions
- Turning and lateral-motion conventions
- Five-leg degraded awareness

WEEK 3. TERRAIN AND CONTACT

- Six-contact estimator and terrain plane
- Depth-assisted footholds with uncertainty
- Incline and uneven-terrain bounds
- Safe operation without camera

WEEK 4. RELIABILITY AND ENERGY

- Fault tree and injected node/leg faults
- Energy per distance and thermal balancing
- Maintainability and spare strategy
- Reliability backlog and regression tests

Evidence package

slow actively leveled 18-DOF hexapod v1.0. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M23. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Book	Introduction to Autonomous Mobile Robots	Locomotion, sensing and mobile-robot integration.	Open
Course	MIT Underactuated multi-contact locomotion	Hybrid/contact reasoning transferable to six legs.	Open
Paper	Hexapod gait generation review search	Compare wave, ripple and tripod gait evidence.	Open
Project	PhantomX ROS packages	Open hexapod descriptions and control examples.	Open
Reference	DYNAMIXEL protocol 2.0	Bus timing, status, errors and multi-actuator operation.	Open
Guide	ROS diagnostics	Health, faults and degraded-mode reporting.	Open

PRACTICE LADDER

- Problem 1 - Scale mechanics and power: Buy six additional actuators only now. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Six-leg gaits: Five-leg degraded awareness. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Terrain and contact: Safe operation without camera. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Reliability and energy: Reliability backlog and regression tests. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

M24. VALIDATION, SYSTEMS ENGINEERING AND PROFESSIONAL RELEASE

MODULE GOAL

Harden the system, close requirements with evidence and present an honest professional portfolio.

VERIFIED ARTIFACT

public v1.0 release, technical report and design defense

Week	Focus	Practical anchor
1	Verification campaign	Requirements-to-test traceability
2	Safety and release	Independent safe-state verification
3	Technical communication	Architecture, timing and safety diagrams
4	Career evidence	Role-specific resume and portfolio index

ACCEPTANCE CRITERIA

- Every major claim has raw evidence
- External reviewer records objections/actions
- No degree/senior/research mastery claim

Time contract

35 hours of theory/reasoning and 40 hours of practical work are the default. Shift up to 8 hours between them only when the artifact and diagnostic checks remain complete.

M24. WEEKLY SYLLABUS AND PRACTICAL EVIDENCE

WEEK 1. VERIFICATION CAMPAIGN

- Requirements-to-test traceability
- 30-50+ trials and uncertainty reporting
- Latency/jitter/current/thermal/energy metrics
- Failure taxonomy and regression suite

WEEK 2. SAFETY AND RELEASE

- Independent safe-state verification
- Fault injection and recovery evidence
- Setup, maintenance and limitations
- Versioned code/CAD/config/data release

WEEK 3. TECHNICAL COMMUNICATION

- Architecture, timing and safety diagrams
- Methods/results/limitations report
- 30-45 minute external design review
- Defend failed approaches and trade-offs

WEEK 4. CAREER EVIDENCE

- Role-specific resume and portfolio index
- Interview questions from actual artifacts
- Open-source contribution or peer feedback
- Post-roadmap specialization plan

Evidence package

public v1.0 release, technical report and design defense. Include setup, assumptions, code/CAD/configuration, raw measurements, metrics, failed cases, limitations and reproduction instructions.

M24. RESOURCES AND PRACTICE PROBLEMS

Type	Book / course / tutorial	How to use it	Link
Handbook	NASA Systems Engineering Handbook	Requirements, verification, validation and reviews.	Open
Guide	Google Engineering Practices	Code review, change quality and maintainability.	Open
Guide	Software testing guide	Test levels, fixtures and systematic regression.	Open
Tool	Sphinx documentation	Versioned technical and API documentation.	Open
Tool	DVC documentation	Version data, models and experiment evidence.	Open
Guide	Jupyter Book	Publish reproducible technical reports and notebooks.	Open

PRACTICE LADDER

- Problem 1 - Verification campaign: Failure taxonomy and regression suite. Submit the result, one quantitative check and one failed or edge case.
- Problem 2 - Safety and release: Versioned code/CAD/config/data release. Submit the result, one quantitative check and one failed or edge case.
- Problem 3 - Technical communication: Defend failed approaches and trade-offs. Submit the result, one quantitative check and one failed or edge case.
- Problem 4 - Career evidence: Post-roadmap specialization plan. Submit the result, one quantitative check and one failed or edge case.

Resource completion rule

Do not consume every resource linearly. Use one primary explanation, one alternative when blocked, and the linked exercises/reference while building the module artifact. A link is supporting material; measured work is the completion evidence.

G1. FOUNDATION AND SAFETY INTEGRATION

PURPOSE

Stop new content, integrate the preceding six modules, repair prerequisite gaps and obtain an external or checklist-based review.

Review area	Required evidence
1	Electronics safety, measurement and E-stop demonstration
2	CAD/mechanics review with load case and prototype
3	MCU timing, bus decoding and fault-state evidence
4	Repair weak prerequisites before actuator work

35-HOUR ALLOCATION

Activity	Hours
Diagnostics and missing prerequisites	7
Cross-module integration	12
Failure tests and rework	9
Review, documentation and decision	7

Gate decision

PASS: continue. CONDITIONAL: continue only with named remediation and owner. FAIL: repeat the weak work; move the calendar instead of deleting later fundamentals.

G2. MODEL, CONTROL AND ROS ARCHITECTURE INTEGRATION

PURPOSE

Stop new content, integrate the preceding six modules, repair prerequisite gaps and obtain an external or checklist-based review.

Review area	Required evidence
1	Actuator model, kinematics, dynamics and control consistency
2	EKF/LQR/QP mathematical diagnostics
3	ROS 2 interfaces, QoS, TF and replay review
4	One-leg HIL demonstration and risk update

35-HOUR ALLOCATION

Activity	Hours
Diagnostics and missing prerequisites	7
Cross-module integration	12
Failure tests and rework	9
Review, documentation and decision	7

Gate decision

PASS: continue. CONDITIONAL: continue only with named remediation and owner. FAIL: repeat the weak work; move the calendar instead of deleting later fundamentals.

G3. AUTONOMY, PERCEPTION AND AI INTEGRATION

PURPOSE

Stop new content, integrate the preceding six modules, repair prerequisite gaps and obtain an external or checklist-based review.

Review area	Required evidence
1	Calibrated perception and SLAM benchmark
2	Planner recovery and manipulation breadth lab
3	ML dataset/model/deployment audit
4	Approve or reject Jetson and learned-locomotion purchases

35-HOUR ALLOCATION

Activity	Hours
Diagnostics and missing prerequisites	7
Cross-module integration	12
Failure tests and rework	9
Review, documentation and decision	7

Gate decision

PASS: continue. CONDITIONAL: continue only with named remediation and owner. FAIL: repeat the weak work; move the calendar instead of deleting later fundamentals.

G4. FINAL VALIDATION AND EXTERNAL DEFENSE

PURPOSE

Stop new content, integrate the preceding six modules, repair prerequisite gaps and obtain an external or checklist-based review.

Review area	Required evidence
1	Close acceptance matrix and statistical report
2	Independently verify E-stop and safe state
3	External design review plus remediation
4	Archive reproducible release and next specialization plan

35-HOUR ALLOCATION

Activity	Hours
Diagnostics and missing prerequisites	7
Cross-module integration	12
Failure tests and rework	9
Review, documentation and decision	7

Gate decision

PASS: continue. CONDITIONAL: continue only with named remediation and owner. FAIL: repeat the weak work; move the calendar instead of deleting later fundamentals.

COVERAGE MATRIX A: FOUNDATIONS AND PHYSICAL ENGINEERING

Topic family	Scheduled block	Depth claim
Programming/Linux/software engineering	M1, M12	Strong implementation depth
Linear algebra/calculus/ODEs	M2	Working foundation
Probability/statistics/uncertainty	M3	Working foundation
Optimization/numerical conditioning	M3, M11, M20	Core plus advanced application
Electronics/signal processing	M4	Working practical depth
CAD/mechanics/FEA/manufacturing	M5	Working practical depth
Embedded/real-time/buses/safety	M6-M7	Strong practical depth
Actuation/motor control/system ID	M7	Strong bench depth

Interpretation

The program covers every common topic family, but specialization depth is intentionally unequal. Legged systems, ROS 2 integration and validation are deeper than manipulation, industrial automation, HRI and research-level AI.

COVERAGE MATRIX B: ROBOTICS CORE

Topic family	Scheduled block	Depth claim
Geometry/FK/IK/Jacobians	M8	Strong practical depth
Rigid-body/contact dynamics	M9, M20	Working plus advanced path
Classical/digital/frequency control	M10	Working practical depth
State space/LQR/Kalman/QP	M11	Working introduction
ROS 2/DDS/TF/ros2_control	M12	Strong implementation depth
Simulation/SIL/PIL/HIL	M13	Strong implementation depth
Vision/depth/point clouds	M14	Working practical depth
Localization/mapping/SLAM	M15	Common-core breadth with benchmark

Interpretation

The program covers every common topic family, but specialization depth is intentionally unequal. Legged systems, ROS 2 integration and validation are deeper than manipulation, industrial automation, HRI and research-level AI.

COVERAGE MATRIX C: AUTONOMY, AI AND PROFESSIONAL CONTEXT

Topic family	Scheduled block	Depth claim
Search/sampling/trajectory planning	M16	Common-core breadth
Task planning/BT/MDP concepts	M16, M21	Working introduction
Manipulation/grasping	M17	Breadth lab
HRI/teleoperation/operator override	M17	Breadth lab
Ethics/privacy/responsible AI	M17	Professional awareness
Cybersecurity/industrial standards	M17	Professional awareness
ML/PyTorch/data/evaluation	M18	Working practical depth
IL/RL/domain randomization/OOD	M21	One bounded experiment

Interpretation

The program covers every common topic family, but specialization depth is intentionally unequal. Legged systems, ROS 2 integration and validation are deeper than manipulation, industrial automation, HRI and research-level AI.

COVERAGE MATRIX D: LEGGED CAPSTONE AND EXTENDED TOPICS

Topic family	Scheduled block	Depth claim
Contact-aided estimation/slip	M19	Strong specialization
Gait/swing/footholds/recovery	M19	Strong specialization
Centroidal/QP/whole-body concepts	M20	Working introduction
MPC/NMPC	M20	Conceptual/optional
Physical quadruped/HIL/sim-to-real	M22	Mandatory capstone
Hexapod leveling/gaits/reliability	M23	Gated physical extension
Validation/statistics/safety/release	M24	Strong evidence depth
Dynamic running/jumping/VLA research	Post-program	Explicitly deferred

Interpretation

The program covers every common topic family, but specialization depth is intentionally unequal. Legged systems, ROS 2 integration and validation are deeper than manipulation, industrial automation, HRI and research-level AI.

ACADEMIC BENCHMARK AND REMAINING DIFFERENCES

Reference program	Common emphasis	Roadmap response / gap
WPI Robotics Engineering MS	Foundations, dynamics, control, robotics electives and capstone	All topic families represented; no faculty grading, transcript or accredited credit.
CMU MRSD	Systems engineering, autonomy, mobility, manipulation/estimation/control, vision, team project and internship	Technical breadth represented; no team studio, business sequence or internship equivalent.
ETH Robotics, Systems and Control	Mechanical/electrical/CS core, modelling, optimization, perception, planning, embedded/distributed systems and AI	Topics represented; roadmap starts below ETH admission prerequisites and is self-directed.

SOURCES

Source	Link
WPI MS	Open official page
CMU MRSD curriculum	Open official page
ETH RSC curriculum	Open official page

Degree boundary

This roadmap is a rigorous self-study and portfolio plan. It cannot reproduce accredited assessment, faculty supervision, peer teams, laboratory access, internship experience or the signalling value of a degree.

QUADRUPED ACCEPTANCE CONTRACT

Area	Minimum evidence
Stand	Stable stand ≥ 60 s on flat floor.
Attitude	Roll/pitch command tracking with RMSE, peak error and settling time.
Incline	Approximately ± 10 -15 degrees only if hardware limits permit; report achieved bound.
Disturbance	Repeatable small push/fixture disturbance with peak deviation, settling and fall rate.
Gait	Controlled slow flat-floor gait with start, stop and turn.
Slip/contact	Foot-slip distance/rate, contact-confidence errors and rejected footholds.
Actuation	Joint/current/temperature/IMU/contact telemetry and enforced limits.
Timing	Control-loop latency/jitter, bus utilization/errors and estimator age.
Safety	Physical E-stop and independently verified safe state.
Statistics	30-50+ trials or statistically justified sample size with raw summaries.

Optional claim

Uneven-terrain traversal is optional. Report terrain dimensions, successful distance, falls, slips, perception dropout and hardware damage-free stopping.

HEXAPOD ACCEPTANCE CONTRACT

Area	Minimum evidence
Scale gate	Quadruped acceptance passes before six additional actuators are purchased.
Stand	Stable stand ≥ 120 s with active body leveling.
Gaits	Wave and ripple mandatory; tripod conditional on stability/power evidence.
Attitude/incline	RMSE, peak and settling; target around ± 10 -15 degrees subject to capability.
Disturbance	Repeatable disturbances with support margin, slip, settling and fall rate.
Reliability	One-node/one-leg fault injection, degraded response and safe stop.
Power/thermal	18-actuator current/temperature, bus load/error rate and battery energy.
Safety/statistics	Physical E-stop, independent safe state and 30-50+ trials or justified sample size.

Claim boundary

The final claim is a slow, actively leveled, contact-aware 18-DOF hexapod. Dynamic running, jumping and research-grade rough-terrain autonomy remain outside scope.

ROLE READINESS AND SPECIALIZATION BOUNDARIES

Target	Evidence expected	Realistic after completion
Junior robotics software / ROS 2	M1, M12-M16, tests, replay and timing	Yes, with strong portfolio and interview practice.
Robot integration / test / simulation	M6-M7, M12-M13, M22-M24	Yes; strongest target.
Entry embedded robotics	M4, M6-M7 plus excellent firmware artifact	Possible; deepen STM32/RTOS for embedded-only roles.
Junior controls / legged robotics	M8-M11, M19-M23 and strong math evidence	Possible but competitive; one specialization should continue.
Perception / ML engineer	M14-M15, M18, M21	Foundation only; add a focused 800-1,200 hours.
Senior/research role	Advanced theory, publications and professional ownership	No; requires years of focused experience.

Application timing

Begin applying after M18 for software/simulation internships or junior roles; apply broadly after M22 when the physical quadruped evidence exists. Do not wait for mastery of every specialization.

POST-PROGRAM ADVANCED SPECIALIZATIONS

Specialization	Additional focused hours	Examples
Advanced nonlinear/robust/MPC control	800-1,200	Lyapunov, robust/adaptive control, NMPC, whole-body control.
State estimation and SLAM	700-1,000	Invariant/error-state filters, factor graphs, visual-inertial SLAM.
Deep perception and robot learning	900-1,400	3D learning, transformers, IL/RL, deployment and research evaluation.
Embedded/real-time motor control	700-1,000	FOC, custom PCB, RT Linux, EtherCAT/CAN-FD and certification.
Mechanical/actuator engineering	700-1,000	Advanced FEA, fatigue, gears, bearings, thermal design and manufacturing.
Manipulation/HRI/industrial robotics	700-1,100	Grasping, force control, collaborative safety, PLC/cell integration.

Mastery rule

Choose one primary and one supporting specialization. Attempting all of them simultaneously produces shallow familiarity, not mastery.

DYNAMIC RUNNING AND JUMPING: WHY IT IS AN EXTENSION

Requirement	Why slow locomotion is not enough	Entry evidence
Actuation	Running/jumping needs high continuous and peak torque, low reflected inertia, impact tolerance and thermal margin.	Measured torque-speed-current-temperature envelope and impact-rated joint stack.
Mechanical design	Repeated landing loads introduce fatigue, gearbox/output-bearing shock and structural resonance.	Drop/impact tests, fatigue load cases and inspected failure modes.
Estimation	Flight phases and intermittent contacts remove stance constraints and amplify timing/bias errors.	Contact-aided estimator remains stable through contact loss and impact.
Control	Hybrid transitions, momentum, force/torque bandwidth and whole-body optimization are substantially harder.	Stable trot/bound in simulation with force limits and classical fallback.
Safety	A dynamic robot can damage hardware or injure a person before a software stop acts.	Remote E-stop, protected test area, tether/gantry and staged energy limits.
Validation	A few successful jumps do not establish robustness.	Repeated trials across payload, voltage, friction and landing error.

OPTIONAL POST-PROGRAM SEQUENCE

- 200-300 hours: upgrade or redesign actuators, joints, power distribution and support rig for impact energy.
- 200-300 hours: hybrid dynamics, centroidal momentum, contact scheduling and whole-body force control.
- 200-400 hours: dynamic simulation, system identification, domain randomization and hardware-in-loop testing.
- 200-400 hours: tethered trot, bound and small-hop progression with independent safety review.
- Expected extension: roughly 800-1,400 focused hours; it is omitted from the core to keep the first physical release realistic and safe.

FINAL ROADMAP VALIDATION CHECKLIST

Validation dimension	Requirement
Hours	24 x 75 h + 4 x 35 h = 1,940 h.
Calendar	20 h baseline plus 22 one-hour recovery sessions; vacations have zero load.
Prerequisites	No electronics, CAD, kinematics or dynamics background assumed.
Common topics	Math, probability, signals, mechanics, embedded, kinematics, dynamics, control, estimation, ROS, simulation, perception, SLAM, planning, manipulation, HRI, ethics/security, ML and integration.
Practical evidence	Every module has a verified artifact and acceptance criteria.
Purchasing	One actuator -> one leg -> 12 actuators -> 18 actuators; Jetson conditional.
Safety	Fusing, current/thermal limits, support rig, physical E-stop and independent safe state.
Statistics	Repeated trials, uncertainty, failure taxonomy and raw summaries.
Language/layout	English only; Arial Regular/Bold; A4 pages; consistent headers/footers.
Claim boundary	No accredited-degree, senior or research-mastery claim; dynamic locomotion is a gated extension.

Completion definition

The roadmap is complete only when the final release can be rebuilt, calibrated, operated, stopped safely and defended using measured evidence by someone other than the author.